

Original vs. Prism-Assisted Paragraph

Part J — AI-Assisted Scientific Writing | Aman Kumar Emaley | MCL2026012

Selected location: Introduction, opening paragraph. The citation numbers are preserved because they refer to the supplied original paper.

Version	Paragraph
A — Original paragraph	Large Language Models (LLMs), enabled by the Transformer architecture, have demonstrated strong capabilities in language understanding and generation [1, 2]. Instruction tuning and reinforcement learning from human feedback have further improved the ability of language models to follow user instructions, but instruction-following should not be equated with factual reliability [3]. As these systems are increasingly applied to scientific tasks, usefulness depends not only on fluency or benchmark performance but also on whether generated statements are factually supported.
B — Prism-assisted paragraph	Large Language Models (LLMs), enabled by the Transformer architecture, have demonstrated strong language-understanding and generation capabilities [1, 2]. Instruction tuning and reinforcement learning from human feedback have improved their ability to follow user instructions; however, instruction-following should not be equated with factual reliability [3]. As these systems are increasingly applied to scientific tasks, their usefulness depends not only on fluency or benchmark performance but also on whether their generated statements are factually supported.
C — Final verified paragraph	Large Language Models (LLMs), enabled by the Transformer architecture, have demonstrated strong language-understanding and generation capabilities [1, 2]. Instruction tuning and reinforcement learning from human feedback have improved their ability to follow user instructions; however, instruction-following should not be equated with factual reliability [3]. As these systems are increasingly applied to scientific tasks, their usefulness depends not only on fluency or benchmark performance but also on whether their generated statements are factually supported.

Brief analysis: Prism improved concision and flow by replacing “strong capabilities in language understanding and generation” with “strong language-understanding and generation capabilities,” adding “however” to clarify the contrast, and making the final subject explicit as “their generated statements.” No citation, technical term, numerical value, causal claim, limitation, or scientific meaning was changed. Version C was accepted as the final verified paragraph because it preserves the original claims and citation placement while improving readability.

Verification: Accepted with a human review. No unsupported claim was introduced.